

NARATHIWAT, Thailand

WMO: 485830

Lat: 6.4271N

Lon: 101.8250E

Elev: 5

StdP: 101.26

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	22.1	22.7	19.8	14.6	25.0	20.5	15.2	25.5	8.0	27.1	6.8	27.4	0.5	190	0.306

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	7.6	33.7	26.5	33.1	26.5	32.5	26.4	27.9	31.9	27.5	31.6	27.2	31.3	3.4	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.9	22.5	30.9	26.2	21.7	30.5	26.0	21.3	30.3	89.5	31.9	87.4	31.8	85.9	31.5	30.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.1	5.4	4.9	DB	20.5	35.6	1.1	1.3	19.7	36.6	19.0	37.4	18.4	38.1	17.6	39.0	(4)
(5)			WB	19.0	29.0	1.3	0.8	18.1	29.5	17.4	30.0	16.6	30.4	15.7	31.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	27.5	26.2	26.6	27.4	28.4	28.7	28.4	28.1	28.0	27.8	27.4	26.6	26.2	(6)
(7)		DBStd	1.26	1.01	0.92	0.95	0.87	0.89	0.81	0.88	0.86	0.77	0.93	1.19	1.07	(7)
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
(10)		CDD10.0	6384	503	464	539	551	580	552	561	559	533	539	499	504	(10)
(11)		CDD18.3	3342	244	231	281	301	322	302	302	301	283	281	249	246	(11)
(12)		CDH23.3	34079	2137	2227	2978	3574	3723	3328	3239	3143	2829	2699	2157	2043	(12)
(13)		CDH26.7	11978	433	575	1018	1494	1612	1381	1289	1250	1055	895	557	418	(13)
(14)	Wind	WSAvg	1.4	1.8	1.9	1.7	1.6	1.4	1.2	1.1	1.3	1.3	1.2	1.1	1.7	(14)
(15)	Precipitation	PrecAvg	2713	218	52	80	64	136	132	148	152	215	302	624	586	(15)
(16)		PrecMax	4016	1011	181	232	146	379	334	244	292	316	550	1243	1299	(16)
(17)		PrecMin	1758	13	9	8	11	41	49	39	56	124	121	308	161	(17)
(18)		PrecStd	651	260	44	68	50	88	69	66	79	60	115	246	320	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.1	31.5	32.6	34.0	34.4	34.1	34.1	34.1	33.5	32.9	32.1	31.2	(19)
(20)			MCWB	25.9	25.5	26.1	26.9	26.8	26.6	26.0	25.6	25.8	25.7	25.9	26.2	(20)
(21)		2%	DB	30.1	30.7	31.7	33.0	33.3	33.1	32.8	32.9	32.4	31.9	31.1	30.2	(21)
(22)			MCWB	25.5	25.4	26.0	26.8	27.0	26.6	26.1	25.9	25.9	25.9	25.9	25.7	(22)
(23)		5%	DB	29.4	30.0	31.1	32.3	32.8	32.4	32.1	32.1	31.7	31.1	30.3	29.6	(23)
(24)			MCWB	25.0	25.2	25.9	26.6	26.9	26.5	26.2	26.0	25.9	25.9	25.8	25.4	(24)
(25)		10%	DB	28.8	29.4	30.7	31.8	32.2	31.8	31.4	31.4	31.0	30.6	29.6	28.9	(25)
(26)			MCWB	24.7	25.0	25.7	26.4	26.7	26.4	26.1	26.0	25.8	25.8	25.6	25.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.9	26.6	27.2	27.9	28.4	28.2	27.7	27.6	27.8	27.9	27.6	27.5	(27)
(28)			MCDWB	29.9	29.9	31.3	32.5	32.7	32.1	31.9	31.5	31.1	30.5	30.0	29.8	(28)
(29)		2%	WB	26.1	26.1	26.7	27.4	27.8	27.6	27.1	27.0	26.8	26.7	26.6	26.3	(29)
(30)			MCDWB	29.3	29.7	30.7	31.9	32.3	31.8	31.3	31.2	30.9	30.5	29.8	29.2	(30)
(31)		5%	WB	25.6	25.7	26.3	27.0	27.4	27.1	26.7	26.6	26.5	26.3	26.1	25.7	(31)
(32)			MCDWB	28.6	29.2	30.2	31.5	31.8	31.4	30.8	30.8	30.6	30.1	29.3	28.6	(32)
(33)		10%	WB	25.1	25.3	26.0	26.7	27.0	26.7	26.4	26.2	26.1	26.0	25.8	25.3	(33)
(34)			MCDWB	28.0	28.7	29.9	30.9	31.4	30.9	30.5	30.4	30.2	29.8	28.8	28.0	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.4	6.1	6.7	7.4	7.6	7.7	7.6	7.8	7.5	6.7	5.5	5.0	(35)
(36)			MCDBR	6.3	7.0	7.4	8.0	8.2	8.3	8.2	8.5	8.3	7.6	7.1	6.4	(36)
(37)			MCWBR	3.0	3.1	3.3	3.5	3.6	3.6	3.5	3.6	3.6	3.4	3.1	3.0	(37)
(38)		5% WB	MCDBR	5.8	6.3	7.0	7.7	7.9	7.9	7.8	8.1	7.8	7.2	6.4	5.7	(38)
(39)			MCWBR	3.1	3.1	3.3	3.6	3.8	3.9	3.7	3.8	3.7	3.5	3.2	3.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.462	0.459	0.465	0.450	0.453	0.466	0.470	0.468	0.461	0.460	0.451	0.468	(40)	
(41)		taud	2.298	2.289	2.290	2.336	2.349	2.334	2.315	2.305	2.319	2.328	2.359	2.295	(41)	
(42)		Ebn at Noon	857	870	863	862	841	819	820	836	854	857	861	843	(42)	
(43)		Edh at Noon	134	138	139	131	126	125	129	133	133	131	125	132	(43)	
(44)	All-Sky Solar Radiation	RadAvg	4.53	5.55	5.80	5.90	5.29	5.07	5.01	5.24	5.15	4.62	3.96	3.79	(44)	
(45)		RadStd	0.35	0.39	0.49	0.26	0.28	0.18	0.28	0.15	0.20	0.31	0.31	0.42	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.20	N/A	N/A	+0.40	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)		+0.17	N/A	N/A	+0.28	+0.22	+0.19	N/A	N/A	+66	+66

Nomenclature: See separate page